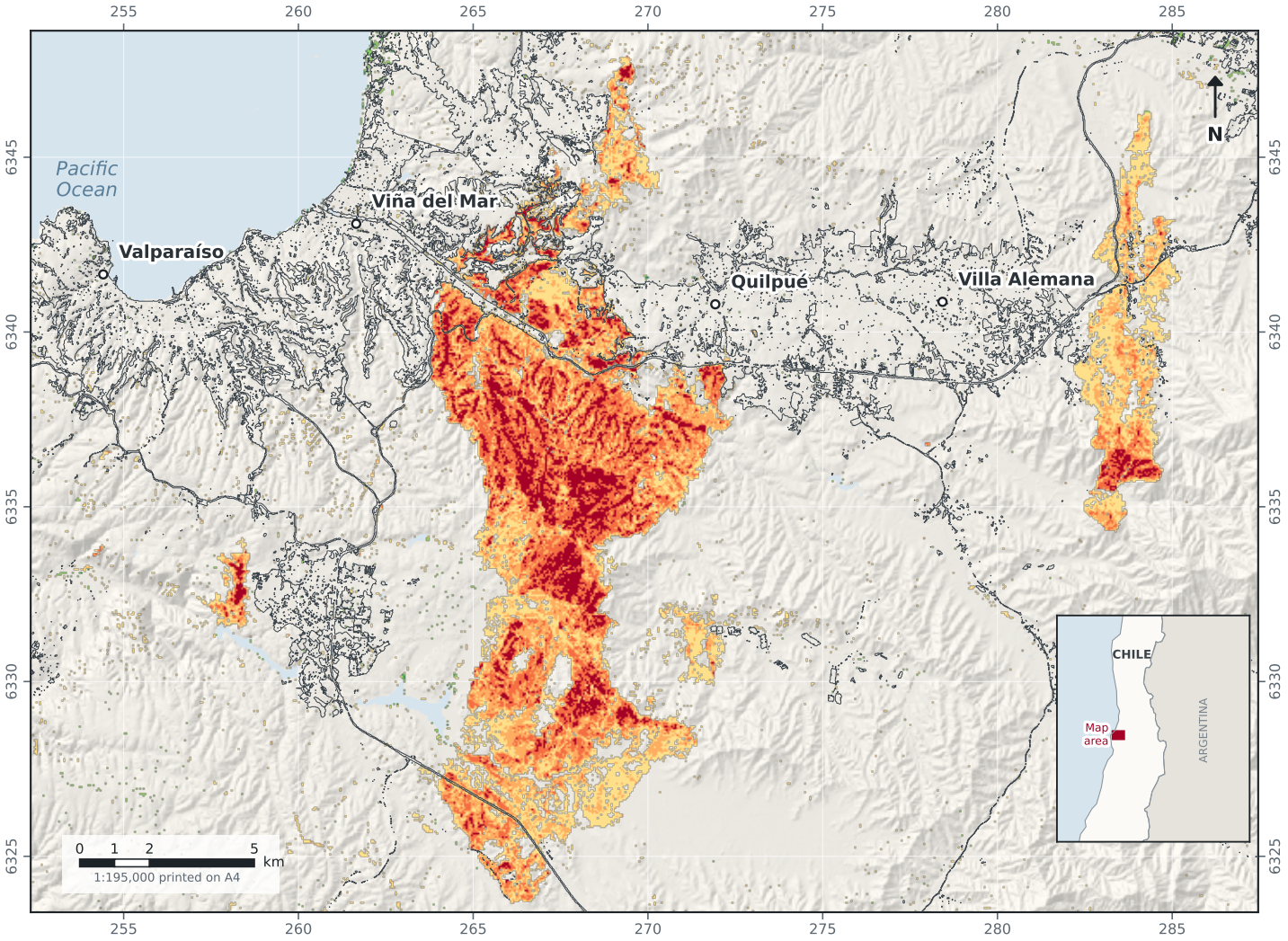
Burn severity of the Viña del Mar fire, February 2024

Valparaíso, Viña del Mar, Quilpué and Villa Alemana, Chile · Sentinel-2 dNBR, validated against Copernicus EMS
Camilo López Barra, geographer

6,759.7 ha burned at moderate-low severity or higher, within a range up to 10,663.0 ha if low severity is included. Copernicus EMSR715 delineated 8,600.5 ha in the same area, inside the range.



SEVERITY (Key and Benson 2006, limits not calibrated for central Chile)

High · Moderate-high · Moderate-low · Low

OTHER CLASSES

Regrowth · Water · Unburned · Built-up 2021

UTM zone 19 S coordinates (km), WGS 84. Cities: GeoNames (CC BY 4.0). Water: Sentinel-2 scene classification. Built-up 2021 (reference, see O5): ESA WorldCover 2021 v200, class 50, © ESA WorldCover project 2021 / Contains modified Copernicus Sentinel data (2021) processed by ESA WorldCover consortium. Relief produced using Copernicus WorldDEM-30 © DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved. Locator: Natural Earth. 20 m pixels.

BURNED AREA BY MUNICIPALITY (ha)

Municipality	Low	Moderate-low	Moderate-high	High	**Moderate-low or higher**	All burned classes
Valparaíso	1,656.2	1,154.4	798.0	330.8	2,283.2	3,939.4
Viña del Mar	865.7	815.9	990.4	589.1	2,395.4	3,261.1
Quilpué	651.0	556.1	595.6	371.0	1,522.6	2,173.6
Villa Alemana	730.4	361.5	140.0	57.0	558.4	1,288.9
Total	**3,903.3**	**2,887.9**	**2,523.8**	**1,347.9**	**6,759.7**	**10,663.0**

VALIDATION IN BRIEF

96.8 %
area-weighted accuracy against EMSR715

11.7 %
commission (25.7 % with low severity)

19.8 %
area-weighted omission (7.0 % with low severity)

54.7 %
of damaged buildings (EMSN188) in any burned class

Contains modified Copernicus Sentinel data 2024. Validation: Copernicus Emergency Management Service (© 2024 European Union), EMSR715 and EMSN188. Relief: Copernicus WorldDEM-30. Built-up: ESA WorldCover 2021. Code MIT, derived data CC BY 4.0. Independent analysis, not endorsed by any of the data providers.

Method and validation

METHOD

1. Scenes: Sentinel-2 L2A S2B_19HBD_20240118 (pre-fire) and S2A_19HBD_20240212 (post-fire) from the Earth Search STAC catalogue; the least cloudy scene inside the study area in each window, a rule fixed before looking at results. Bands B8A, B12 and scene classification, 20 m.
2. Reflectance: the +1000 digital-number offset is measured on water pixels instead of taken from the catalogue metadata (see O4).
3. Quality mask: no data, saturated, dark, shadow, cloud, cirrus, water and snow removed from the pre-fire scene; dark pixels and shadows kept in the post-fire scene (burn scars are dark).
4. dNBR with offset: NBR = (NIR - SWIR) / (NIR + SWIR); dNBR = NBR pre - NBR post, minus the mean dNBR of an unburned ring 500 m to 2 km around the preliminary burned area (Parks et al. 2018).
5. Classes: Key and Benson (2006), dNBR x 1000: low 100 to 270, moderate-low 270 to 440, moderate-high 440 to 660, high 660 or more; unburned -100 to 100.
6. Reproducibility: the content hash of the class raster (ef09b76717fc…) is identical when rebuilt from scratch on Windows and Linux.

VALIDATION (PROTOCOL AND HEADLINE RULE FIXED BEFORE MEASURING)

M1, omission in the urban interface: 300 of the 7,270 building damage points of Copernicus EMSN188 (Pléiades Neo, 0.3 m, 'Las Tablas' area), drawn with seed 20240202 and at least 100 m apart. 54.7 % fall in any burned class (Wilson 95 %: 49.0 to 60.2); 14.0 % at moderate-low or higher.

M2, burned / unburned against the EMSR715 delineation v2 (WorldView-2, 2 m), inside the WorldView-2 footprint (69,926.0 ha; 8,600.5 ha burned in the reference). 300 points per predicted stratum, at least 200 m apart; accuracy and omission weighted by stratum area (Olofsson et al. 2014).

Threshold	Predicted burned (ha)	Area-weighted accuracy	Commission (Wilson 95 %)	Area-weighted omission
Moderate-low or higher	6,759.7	96.8 %	11.7 % (8.5 to 15.8)	19.8 %
All burned classes	10,663.0	95.2 %	25.7 % (21.1 to 30.9)	7.0 %

Headline rule: publish the threshold with the higher area-weighted accuracy and the other as the range. Moderate-low or higher won by 1.6 points. EMSR715 used Sentinel-2 as its pre-event reference, so it is mostly, not fully, independent of this map.

LIMITATIONS

O1. Vegetation, not buildings. About 45 % of the buildings mapped as damaged by EMSN188 fall outside every burned class: dNBR at 20 m measures loss of vegetation, and building damage needs very high resolution imagery.

O2. Low severity adds commission. Including it raises commission from 11.7 % to 25.7 %; leaving it out raises omission from 7.0 % to 19.8 %. Hence a range.

O3. Uncalibrated classes. Key and Benson limits were defined for North American ecosystems; no field severity plots were available for central Chile.

O4. Catalogue offset. The Earth Search COGs of these scenes do not carry the +1000 offset their STAC metadata declares; applying it roughly doubles the burned area.

O5. Built-up 2021 is a reference outline (ESA WorldCover class 50: buildings, roads, structures); no figure uses it. It misses later settlements and can under-represent scattered self-built housing.

REPRODUCE

Notebook 01 rebuilds the map from scratch and checks its hash; notebook 02 recomputes M1 and M2; severidad.contexto rebuilds the context layers of the map. Details in README.md.

REFERENCES

Key, C. H., and Benson, N. C. (2006). Landscape assessment (LA): sampling and analysis methods. FIREMON. USDA Forest Service, RMRS-GTR-164-CD.

Olofsson, P., et al. (2014). Good practices for estimating area and assessing accuracy of land change. Remote Sensing of Environment, 148, 42-57.

Parks, S. A., et al. (2018). Mean composite fire severity metrics computed with Google Earth Engine offer improved accuracy and expanded mapping potential. Remote Sensing, 10(6), 879.

Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. Journal of the American Statistical Association, 22(158), 209-212.

Zanaga, D., et al. (2022). ESA WorldCover 10 m 2021 v200. Zenodo. doi:10.5281/zenodo.7254221.

Contains modified Copernicus Sentinel data 2024. Validation: Copernicus Emergency Management Service (© 2024 European Union), EMSR715 and EMSN188. Relief: Copernicus WorldDEM-30. Built-up: ESA WorldCover 2021. Code MIT, derived data CC BY 4.0. Independent analysis, not endorsed by any of the data providers.